

Yash Verma

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EDUCATION

- **Indian Institute of Information Technology, Lucknow** Lucknow
Bachelor of Information Technology; CGPA: 8.32 Apr 2020 - Apr 2024
- **Relevant Courses:** Data Structures & Algorithms, Operating System, System Design, DBMS, Object Oriented Programming, Distributed Systems, Software Management, Design Patterns.

EXPERIENCE

- **Browerstack** Mumbai (Mallard West)
SDE - Backend - Ruby, Rails, Express, JavaScript, SpringBoot, Java Apr 2025 – Present

Product Features Shipped

- Shipped an enterprise-grade **RBAC access-control system** (roles, hierarchies, **14 check-points**), enabling organisation-wide permission management for **50k+** enterprise customers.
- Delivered **multi-language SDK support** across **5B+** API calls via **client-aware routing**, expanding platform reach to non-English enterprise markets.
- Consolidated **150M test runs and re-runs** into **accurate aggregate results** (status, duration), eliminating duplicate entries.
- Built **CDC pipeline** from **MySQL to ClickHouse** for a **10TB** database, supporting **5B+ inserts/updates** for real-time analytics. Along with **IAC provisioning** for **35 Debezium CDC connectors** with auto-sync, eliminating manual config drift across 4 regions.
- Launched BrowserStack's **Australia AWS region** to meet **GRR/GDPR**, now handling **360K requests/hour** in production.
- Halved per-session auth calls (**2 → 1**) for accessibility products via a lightweight config service, cutting **102K auth calls/day**.

Platform & Performance

- Drove **208× query speed up** (9.6s → 46ms) on a **524M-row analytics table** via zero-downtime schema migrations across 4 regions.
- Cut dashboard load from **4 min → 3s** by **indexing high-cardinality queries**, directly unblocking a stalled enterprise deal.
- Executed **zero-downtime PK migrations** on 3 tables of **1B+ rows** (~1 TB, **2.9M rows/day**) to prevent production outages.
- Offloaded heavy aggregate queries from MySQL to **ClickHouse**, **stabilizing database CPU from 96%** and enabling future growth.
- Remediated **SQL injection** across 200+ APIs and shipped **Static CI security scanning** covering **~98% of production traffic**.
- Resolved **2 critical vulnerabilities** (RCE and SSRF bypass) and enabled **TLS for database connections** to harden production security.

- **Zus (formerly 0chain)** Cupertino, CA (Remote)
Software Development Engineer (Freelance) - Golang, GitHub Actions, Gin, Grafana, Helm, Kafka Apr 2024 – Mar 2025

- Spearheaded the **UAT test suite** for **enterprise storage**, reducing production bugs by **90%** and significantly improving release quality.
- Refactored and migrated **GoSDK**, reducing app size by **50%**, while assisting cross-platform teams in the transition.
- Benchmarked **S3-compatible storage** for **500 million** bulk files and **40 TB** large files, achieving sub-**100ms** latency on both **Enterprise** and **Standard** allocations.
- **Load-tested** streaming videos using **JMeter**, optimizing performance based on varying data sets and **Erasur Coding (EC) parity nodes** for efficient allocation.
- Developed **automated workflows** for **view change** and dynamic removal of **Sharders/Miners**, boosting developer efficiency by over **90%**.
- Built an **automated alert system** for **missing network blocks** and integrated **build tags** into Slack using **Python scripts, BeautifulSoup, Webhooks, and Block Kit**.
- Enhanced **multi-wallet support** to improve **concurrency handling** in **load tests**, while refactoring legacy load test code with a unified **GoSDK** for streamlined performance.

- **Zus (formerly 0chain)** Cupertino, CA (Remote)
Android Engineer (Freelance) - Kotlin, Java, AndroidSDK, GitHub Actions Feb 2022 – Mar 2024

- Architected two Android apps, **Bolt** (crypto wallet) and **Vult** (decentralized storage platform), increasing the Android user base by **20%** for decentralized storage using **ZCN crypto**.
- Led and mentored two junior developers, enhancing **code quality** by **40%** and ensuring **95%** on-time delivery through effective **Scrum leadership**.
- Designed and implemented **release pipelines** for **Android apps**, reducing manual effort and accelerating release cycles by **70%**.
- Developed an **Android app** with **Split Key Functionality** to act as an intermediary between the **app** and the **blockchain**, enabling double-signing of transactions via **Bluetooth**.

- **Google Summer of Code Oppia Foundation** Remote
GSOC - Kotlin, GitHub Actions, JUnit, UIAutomator, Gradle, Bazel, AndroidSDK Jun 2021 – Sept 2022

- Designed and architected a robust **CI/CD infrastructure** to streamline build processes and enhance deployment efficiency.

- Implemented automated build triggers for **alpha**, **beta**, and **test** releases, seamlessly signing and uploading binaries to **Google Drive**, **GCP Bucket**, and **Google Play Store** tracks.
- Developed an intelligent **cherry-picking workflow** that autonomously merges selected changes into the appropriate **flavor branch**.
- Engineered an automated logging system that tracks each build and release step, updating **GitHub issues** with workflow triggers and contextual comments.
- Spearheaded **Android Release Automation**, integrating streamlined processes for **signing**, **building**, **testing**, and **publishing** applications to Google Play.
- Authored custom **Kotlin scripts** to simplify essential tasks such as generating **AAB bundles**, securely **signing** artifacts, and automating **Google Play** uploads.
- Leveraged **GitHub Actions** in conjunction with custom **Kotlin scripts** executed via **Bazel** to optimize **CD pipelines** for **alpha**, **beta**, and **nightly releases** on scheduled **cron jobs**.

● **Oppia Foundation Github Link** [Link to Merged PRs]

Mountain View, CA (Remote)

Open Source Contributor Android

June 2021 – April 2022

- Contributed to the migration of the build system from **Gradle** to **Bazel**, achieving a **10x** improvement in build speeds.
- Developed modular **fragments** encapsulated within standalone **activities**, ensuring seamless functionality across mobile and tablet devices with support for **multiple screen sizes**.
- Implemented **MVVM** (Model-View-ViewModel) and **MVP** (Model-View-Presenter) architectures to structure and optimize all activities and fragments.
- Authored over **70+** Unit Tests, **20+** Instrumentation, and End-to-End tests using **Espresso**, **Robolectric**, and **UIAutomator**, with **JUnit** as the test runner for Android.
- Mentored and onboarded **3+** students in the **Android repository**, guiding them to become active **collaborators**.
- Enhanced project documentation by creating a comprehensive **GitHub Wiki** on **Protobuf**, significantly improving understanding and adoption of the **Protobuf** library within the repository.

● **Gnome Foundation**

Remote

Open Source Contributor - Tech Stack: C, XML, GTK+

Sept 2021 – Feb 2022

- Added a new feature in the **path-bar** of **Nautilus file manager**, improving **user efficiency** by **20%** by enabling seamless opening of folders and files.
- Designed and implemented a new button using **XML**, enhancing **user interaction** by **15%** through improved interface functionality.
- Refactored and optimized **UI components** in Nautilus, reducing render time by **30%** and improving overall navigation fluidity.
- Implemented keyboard shortcuts for **quick folder navigation**, enhancing accessibility and workflow efficiency.
- Integrated **drag-and-drop support** for seamless file organization, significantly improving **user experience** in Nautilus.

● **BSHAPP**

Andhra Pradesh

Android Freelance Developer

May 2021 – Jun 2021

- Added support for **multiple languages**, expanding the app's user base by **30%** and enhancing accessibility in diverse markets.
- Implemented **Jetpack Navigation** with single activity and multi-fragment architecture, improving app performance by **25%**.
- Integrated **REST API** with **Retrofit**, reducing data retrieval time by **20%** and improving HTTP request efficiency.
- Optimized **network caching** mechanisms using **OkHttp**, significantly reducing redundant API calls and enhancing data load speeds.
- Refactored legacy codebase to adopt **Kotlin Coroutines** for asynchronous processing, leading to a **40%** improvement in app responsiveness.
- Enhanced security by implementing **encrypted shared preferences** and **biometric authentication**, strengthening data protection and user privacy.

PROJECTS

- **Parking Lot Management System:** Developed a RESTful API using **Node.js**, **Express.js**, and **MongoDB** to manage parking spaces. Integrated **Winston** for logging and conducted unit and integration testing with **Jest**.
- **Airline ticketing system:** Built using **Spring Boot** in **MVC architecture** where a user can book tickets, query flights and cancel tickets also **Dockerized**.
- **Trip Feedback System:** Designed a feedback system for **Travelers** and **Transport Managers** with secure feedback submission, user authentication, using **Spring Framework**.
- **Festify:** Hackathon and event management multi-module Android app with NodeJs backend and MongoDB database.
- **Hello World:** Android application showcasing **OChain** blockchain storage SDK, designed in Kotlin.
- **Wake-Lock:** Runs a sticky service in the background which adds a partial wake lock to your screen.
- **Automated Window Manager:** A CLI tool to automate window management in Linux.
- **Backup System Ubuntu:** A bash script to back up apt, snap, and other config files.
- **GUI-Automator:** Python script for downloading, sorting, and extracting files from Google Drive.
- **What's App Clone:** Android chat app featuring Google and email authentication.
- **Video Chatting:** Android app for video conferencing using Jitsi Meet API with a secure server setup.
- **My Resume:** Automated deployment of resume updates using **GitHub Actions** and **GCP** to Google Drive.

ACHIEVEMENTS

- Practised over **700+** DSA problems on Leetcode.

- Member of **Oppia Foundation** for **2 years**.
- Completed **Cloud Engineering tracks** 30 Days of Google Cloud Program.

TECHNICAL SKILLS

- **Programming & Scripting:** Java, Go, Kotlin, Python, JavaScript, SQL, Bash, C/C++
- **Backend Frameworks:** Spring Boot, Gin, Express.js, REST/GraphQL APIs, RxJava, Dagger/Hilt
- **DevOps & Cloud:** Docker, Kubernetes, GitHub Actions, CI/CD, GCP, Linux, Helm, Prometheus/Grafana/Loki
- **Databases & Testing:** MySQL, MongoDB, PostgreSQL, Firebase, JUnit, Mockito, UIAutomator
- **Tools & Libraries:** Git, Retrofit/OkHttp, Gradle, Bazel, Android SDK (cross-platform support)

SOFT SKILLS

- **Communication:** Excellent verbal and written communication skills, effective team collaboration.
- **Leadership:** Strong leadership skills, mentoring and guiding junior developers.
- **Problem Solving:** Analytical and critical thinking, adept at problem-solving and debugging.
- **Adaptability:** Ability to adapt to new technologies and work in a fast-paced environment.

HOBBIES

- **Open Source:** Contributing to open-source projects, mentoring new developers.
- **Programming:** Solving DSA problems on Leetcode, Codeforces, and Codechef.
- **Reading:** Reading tech blogs, articles, and books on software development.
- **Music:** Playing guitar, listening to music, and attending live concerts.
- **Electronics:** Building electronic circuits, transformers, etc.